Harrison Felstead

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PROFESSIONAL SUMMARY

Final-year Bachelor of Engineering (Honours) (Mechatronics) and Bachelor of Science (Computer Science) student at Monash University, completing June 2027. Strengths in AI and neural network development, PLC and embedded control systems, and integrating hardware with software. Final year project converts foot pressure data into 3D-printed custom insoles, validated on a Pedar pressure system. Experienced in leading small teams and delivering working prototypes. Seeking graduate and internship roles in AI and engineering development.

AREAS OF EXPERTISE

- **AI development:** neural networks (PyTorch), unsupervised learning, computer vision (YOLO)
- **PLC control systems:** ladder logic and HMI (CODESYS), ISA-95, SCADA concepts
- **Hardware and software integration:** STM32, Arduino, Raspberry Pi
- **Testing, troubleshooting and verification** of robots and microprocessor systems
- **App development:** PostgreSQL, Expo, React, Tailwind
- **Data organisation**, process streamlining and technical documentation
- **CAD and additive manufacturing:** SolidWorks (CSWA), TPU 3D printing
- **Team leadership** and project delivery in teams of 3 to 5

PROFESSIONAL EXPERIENCE

ENGINEERING & TECHNICAL PROJECTS

Final Year Project: Custom Orthopaedic Sole Creation using Unsupervised Learning *2026 – Present*

Monash University · Team of 3 (only Mechatronics student)

- Developed an end-to-end workflow from raw plantar pressure data to a 3D-printable customised sole, using unsupervised learning (K-means) to group regions of similar pressure.
- Printed 5 TPU prototypes and validated them on a Pedar in-shoe pressure system, measuring pressure change across the gait cycle.
- Corrected incorrect material allocation on the pressure map by adding parameters and methods that follow accepted orthotic design practice, giving a significant improvement.
- With no existing quantitative benchmark, defined our own comparison measures; results indicate improvement without claiming a complete definition.
- Managed scope by cutting feature additions to focus on pressure reduction. Applied technical documentation, test planning and iteration from earlier mistakes.

Carpark Carousel PLC System *Sem 2 2026*

Monash University · Team of 5

- Designed a multi-PLC control system for a carpark carousel using CODESYS (ladder logic and HMI) and built a small prototype for demonstration, applying ISA-95 and SCADA concepts.
- Designed the safety PLC and the communication between PLCs, delivering dynamic fault detection and working multi-PLC communication.

Fruit-Finding Robot: Pathfinding and Vision System *Sem 2 2025*

Monash University · Team of 4

- Developed A* pathfinding and a YOLO vision system in Python so a robot could identify fruit and navigate around obstacles using only what it could see.
- Programmed and implemented the vision system and optimised parameters during testing.
- Robot collected all fruit within the time limit with no obstacle collisions.

NCCD Mobile App and Website *Early 2026 – Present*

Client project (outside university)

- Built a data logging system that removes roadblocks for teachers and school staff collecting the data used to support school funding, using PostgreSQL, Expo, React and Tailwind.
- Learned the fundamentals of regulated data storage, UI/UX design, and working with clients to deliver a product.
- Prototypes are performing well; a school pilot is scheduled for Term 1 2027.

Research & Development Team Member *Apr 2026 – Sep 2026*

Monash BrewLab, Monash University

- Researched new recipes and adapted them to our lab conditions and equipment, then produced them.
- Trained most new members to use the space and built their confidence with the equipment.

LEADERSHIP

Brewkeepers Lead *2026 – Present*

Monash BrewLab, Monash University · Sub-team of 3 to 5

- Lead the sub-team responsible for lab upkeep, logistics and administration, keeping operations organised so the wider team can run its work.

Education Officer *Sep 2024 – Sep 2025*

Monash Engineering Students Society (MESS), Monash University

- Designed and wrote the "Ed Guide", a first-year resource adopted by the Engineering faculty.
- Mentored first-year Engineering students.

Vice President *Sep 2024 – Present*

German Liechtenstein Austrian Swiss Society (GLASS), Monash University

- Oversee committee operations, guide decision-making and plan the society's annual program, including cross-cultural events.

Wellbeing & Equity Officer *Sep 2023 – Sep 2024*

Monash Engineering Students Society (MESS), Monash University

- Designed and launched mental health awareness and inclusivity programs across the Engineering faculty.

OTHER EMPLOYMENT

OSHC Educator *Summer 2025 – 2026*

Holy Cross Primary School, Cairns, QLD

- Supervised and guided primary school students in structured activities, building the communication and patience needed for client-facing work.

EDUCATION AND QUALIFICATIONS

Bachelor of Engineering (Honours) (Mechatronics) / Bachelor of Science (Computer Science) *2022 – Present*

Monash University, Melbourne, VIC · Expected completion June 2027

- Coursework: neural networks and machine learning (PyTorch), control systems, PLC systems, signal processing, embedded systems (C/C++), robotics, CAD.
- Safety training: chemistry units covering risk assessments, PPE, and safe chemical handling and disposal.

Queensland Certificate of Education *2018 – 2021*

St Augustine's Marist College, Cairns, QLD · ATAR: 97.80

- **Certified SOLIDWORKS Associate (CSWA)**
- **Google Project Management Certificate**, Coursera, Dec 2023
- **Google IT Support Certificate**, Coursera, Dec 2023
- **Working with Children Check** held

CONTINUING PROFESSIONAL DEVELOPMENT

- Anthropic: AI Fluency (Framework and Foundations; for Builders)
- Anthropic: Claude Code 101 and Claude Code in Action
- Anthropic: Introduction to Agent Skills and Subagents
- Anthropic: Model Context Protocol (Introduction and Advanced Topics); Building with Claude API

SKILLS

Programming: Python, C++, C, Assembly, MATLAB
AI and Data: PyTorch, YOLO, K-means, Jupyter
Web and App: React, Expo, Tailwind, PostgreSQL
Hardware and CAD: CODESYS, STM32, Arduino, Raspberry Pi, SolidWorks
Tools: Git, Linux, Microsoft Office Suite, Agile / Scrum
Languages: English (Native), German (A2)

INTERESTS

Surfing, hiking, and building quality-of-life apps. Volunteer, Rosies Friends on the Street, Cairns (2014 – 2017), preparing and distributing food to vulnerable community members.

REFERENCES

Available upon request.